ZACHARY L. STRASBERG
Department of Earth and Planetary Sciences
University of New Mexico
Northrop Hall, 221 Yale Blvd NE
Albuquerque, NM 87131
Email: zstrasberg@unm.edu, Phone: (571) 226-7221

EDUCATION

2024-Present — ***Ph.D., Earth and Planetary Sciences***, University of New Mexico, Department of Earth and Planetary Sciences (Anticipated May 2027)

- Primary Project: Enhanced Prediction In Chaotic Systems with Physics-Learned Autonomous Neural Networks (EPIC-PLANN)
- Secondary Project: Surface Ocean CO_2 Atlas Observing System Simulation Experiment in Large Marine Ecosystems
- Advisor: Dr. Louis A. Scuderi
- Committee Members: Dr. Joe Galewsky, Dr. Nicole Jackson, Dr. Jonathan Sharp

2021–2024 — ***M.S., Earth and Planetary Sciences***, University of New Mexico, Department of Earth and Planetary Sciences

- Thesis: Death by a Thousand Paper Cuts: Climate Change and Drought in the Western United States
- Advisors: Dr. Louis A. Scuderi, Dr. Yemane Asmerom
- Committee Member: Dr. Joe Galewsky
- GPA: 4.28/4.33

2017–2021 — ***B.S., Geology***, James Madison University, Department of Geology and Environmental Science

- Minors: Statistics, Mathematics; Honors College
- GPA: 3.88/4.00 – Magna Cum Laude
- Thesis: Quantifying Late Oligocene Seasonal Precipitation in South China with Fossil Evergreen Wood
- Advisor: Dr. William E. Lukens

RESEARCH INTERESTS

Drought analysis and prediction, machine learning, paleoclimatology, climate modelling, statistics, ocean acidification, dendroclimatology, geochemistry, stable isotopes, karst

AWARDS AND SCHOLARSHIPS

2021–2022	***EEE/Black/Gorham Fellowship for Top Incoming Earth and Planetary Sciences Graduate Student***, University of New Mexico
2021	***Dean's Award for Outstanding Senior in Geology (GPA)***, James Madison University
2019	***William Frangos Memorial Endowment for Geophysics and Quantitative Geology Research***, James Madison University
2018	***Catherine King Frazier Scholarship for Most Promising Geology Freshman***, James Madison University
2017–2020	***President's List*** (4x) and ***Dean's List*** (4x), James Madison University
2017–2021	***Second Century Three-Quarter Tuition Scholarship***, James Madison University

RESEARCH EXPERIENCE

2024-Present ***Research Assistant (Ph.D.)***: University of New Mexico
- Trains, validates, and tests sub-seasonal to seasonal (S2S) scale precipitation and drought prediction machine learning models.
- Compiles and generates synthetic and weather data from NCAR/USGS's CONUS404 and ECMWF's ERA5 reanalysis datasets.
- Collaborates with Sandia National Laboratories Laboratory Directed Research and Development (LDRD) team on the Enhanced Prediction In Chaotic Systems with Physics-Learned Autonomous Neural Networks (EPIC-PLANN) project. Funded by Sandia National Laboratories Laboratory Directed R&D (LDRD) 24-0128
- Advisor: Dr. Louis A. Scuderi

2024-Present ***Faculty Assistant:*** University of Maryland/NOAA NCEI NESDIS
- Updates a MySQL/MariaDB relational database with logistical information about vessels in NOAA's Ship Cruise Catalog.
- Advisors: Dr. Hyelim Yoo, Dr. Christopher Paver, Dr. Liqing Jiang

2021–2024 ***Research Assistant (M.S.)***: University of New Mexico
- Analyzed historical drought datasets to make interpretations about causes, history, and impacts of droughts in the Western United States in the context of climate zones.
- Used machine learning methods to model the impacts of increasing temperatures on drought severity in the Western US.
- Advisors: Dr. Louis A. Scuderi, Dr. Yemane Asmerom

2023 ***Lapenta Intern***: William M. Lapenta Student Internship Program: NOAA Ocean Acidification Program (OAP).
- Performed observing system simulation experiment (OSSE) of Surface Ocean CO_2 Atlas (SOCAT) pCO_2 RandomForestRegressor machine learning model.
- Developed tools that assist OAP staff in tracking spatial and temporal coverage of SOCAT pCO_2 measurements in Large Marine Ecosystems (LMEs).
- Advisors: Dr. Louis A. Scuderi, Dr. Yemane Asmerom

2020–2021 ***Honors Undergraduate Researcher***: James Madison University
- Statistically analyzed paleoclimate proxy data to modern Global Historical Climatological Network (GHCN) weather station data to make interpretations about the East Asian Monsoon in Oligocene Southern China.
- Extracted fossil wood samples for $\delta^{13}C$ analysis.
- Advisor: Dr. William E. Lukens

2019–2021 ***Volunteer Researcher***: Florence Bascom Geoscience Center: US Geological Survey
- Collected and processed loss-on-ignition (LOI) data from southern Virginia peat core for palynological study.
- Advisors: Dr. Miriam Jones, Kristen Hoefke

2019–2020 ***Undergraduate Researcher*****:** James Madison University
- Collected, prepared, and processed $\delta^{13}C$ and $\delta^{18}O$ samples from a western Virginia stalagmite to make interpretations about past environmental changes.
- Advisor: Dr. Ángel A. Garcia

2018 ***Summer Intern*****:** George Mason University
- Contributed to the publication of a paper about changes to the South Atlantic Subtropical High-pressure system with CMIP5 model data.
- Performed statistical analyses in GrADS supported by a comprehensive literature review.
- Advisors: Dr. Natalie J. Burls, Dr. Abdullah al-Fahad

2016–2017 ***Aspiring Scientists Summer Intern*:** George Mason University
- Completed two group research projects focused on the relationship between climate model variability and predicted changes due to global warming.
- Advisors: Dr. Natalie J. Burls, Dr. Chris Selman, Dr. Abdullah al-Fahad

PEER-REVIEWED PUBLICATIONS

Strasberg Z.L., Scuderi, L.A. (In Preparation). Death by a Thousand Paper Cuts: Climate Change and Drought in the Western United States.

Polyak, V.J., Asmerom, Y., Curry, B., Lavery, D., **Strasberg, Z.L.,** Cutler, S., Song, W., Crossey, L.J., Karlstrom, K. (2023). Large negative δ^{238}U anomalies in speleothems and travertines: Indicators of deep-seated endogenic sources of CO_2 and H_2S: *Geology*, vol. 51 (11), p. 1048-1052.

Fahad A.A., Burls N.J., **Strasberg Z.L.** (2020). How will Southern Hemisphere subtropical anticyclones respond to global warming? – Mechanisms and seasonality in CMIP5 and CMIP6 model projections: *Climate Dynamics*, vol. 55, p. 703-718.

ABSTRACTS AND PRESENTATIONS

17. **Strasberg Z.L.**, Chapman, G.W., Teeter, C., Mott, J., Scuderi, L.A., Jackson, N.D. (2024). Developing Machine-Learning and Statistical Models for Western US Seasonal Precipitation. Abstract *#A51L-1838. Poster Presentation*. American Geophysical Union Fall Meeting 2024, Washington, DC.
16. Jackson, N. D., Chapman, G.W., Teeter, C., Mott, J., Scuderi, L.A., **Strasberg, Z.L.** (2024). Machine Learning and Reservoir Computing for Watershed-Scale Drought Prediction American Geophysical Union Abstract *#H11E-03. Oral Presentation*. American Geophysical Union Fall Meeting 2024, Washington, DC.
15. **Strasberg Z.L.**, Scuderi, L.A. (2024). Death by a Thousand Paper Cuts: Climate Change and Drought in the Western United States. *Poster Presentation*. Presented at National Center for Atmospheric Research Earth System Predictability Across Timescale Workshop, April 2024.
14. **Strasberg Z.L.**, Scuderi, L.A. (2024). Death by a Thousand Paper Cuts: Climate Change and Drought in the Western United States. *Poster Presentation*. Presented at University of New Mexico Earth and Planetary Sciences Graduate Research Symposium, April 2024.
13. **Strasberg Z.L.**, Scuderi, L.A. (2024). Death by a Thousand Paper Cuts: Climate Change and Drought in the Western United States. Presented as M.S. Defense, University of New Mexico Earth and Planetary Sciences Department, February 2024.
12. **Strasberg Z.L.**, Scuderi, L.A. (2023). Death by a Thousand Paper Cuts: Climate Change and Drought in the Western United States. American Geophysical Union Abstract *#GC41K-1239. Poster Presentation*. American Geophysical Union Fall Meeting 2023, San Francisco, CA.
11. **Strasberg Z.L.**, Sharp J.D., Goldsmith K., Perotti E.A., Gledhill, D.K., Jiang, L. (2023). Surface Ocean CO_2 Atlas Coverage and Observing System Simulation Experiment in Large Marine Ecosystems: American Geophysical Union Abstract *#OS41C-1764. Poster Presentation*. American Geophysical Union Fall Meeting 2023, San Francisco, CA.
10. Simpson, A., **Strasberg Z.L.**, Scuderi, L.A. (2023). Drought Quantification Methods and the 21st Century Drought in the Western United States. Presentation at Undergraduate Research Opportunity Conference by A. Simpson on April 21, 2023
9. **Strasberg Z.L.** (2022). How is the current Western United States drought different than previous droughts? – An analysis of weather, tree-ring, and speleothem records from the past 1,200 years. Presented at the University of New Mexico as an M.S. Comprehensive Exam.
8. **Strasberg Z.L.** (2021). Placing late Oligocene paleoclimate estimates for southern China into a modern climate framework. Presented at James Madison University Research Symposium.
7. **Strasberg Z.L.**, Lukens W.E., Schubert B.S. (2021). Applying spatial analog matching to discern late Oligocene paleo-monsoon climatology in southern China: Geological Society of America

Southeastern Section – 70th Annual Meeting. *Oral Presentation.* vol. 53, no. 2, session no. 5. Abstract # 362306

6. **Strasberg Z.L.**, Lukens W.E., Schubert B.S. (2020). Placing late Oligocene paleoclimate estimates for southern China into a modern climate framework: American Geophysical Union Fall Meeting. *Poster Presentation.* Poster # PP024-012. Abstract # 750466
5. **Strasberg Z.L.,** Garcia Jr. A.A. (2020). Paleoenvironmental Reconstruction of Owl Cave, Highland Co., VA Using $\delta^{13}C$ from Stalagmites as a Proxy, Geological Society of America *Abstracts with Programs.* vol. 52, no. 2.
4. Maslock C.L., Rice K.T., **Strasberg Z.L.**, Ranieri C.M., Sexton F.P., Washington D.M., Williams S.N. (2020). Reconstructing Late Pleistocene North Pacific Ice Rafting History Through Two Glacial Cycles, Geological Society of America *Abstracts with Programs.* vol. 52, no. 2.
3. Fahad A.A., **Strasberg Z.L.**, Burls N.J. (2018). How will the South Atlantic Subtropical High Respond to Climate Change? – Evaluating CMIP5 Model Predictions. *Poster Presentation.* American Geophysical Union Fall Meeting, Washington, D.C.
2. **Strasberg Z.L.**, Chon R., Fahad A.A., Burls N.J. (2017). How will the South Atlantic Anticyclone Respond to Climate Change? -- Evaluating CMIP5 Model Predictions. Presented at Aspiring Scientists Summer Internship Program (ASSIP) Poster Session, George Mason University, Manassas, VA.
1. **Strasberg Z.L.,** Gross M., Selman C., Kodama K., Burls N.J. (2016). Using Satellite Radiation Data to Identify Sources of Model Sea Surface Temperature Bias. Presented at ASSIP Poster Session, George Mason University, Manassas, VA.

RELEVANT WORK EXPERIENCE

2015–2023 ***Data Analyst***: ADI Compliance Consulting, Fairfax, VA

- Provided technical support to senior economist to evaluate lender compliance with the Equal Credit Opportunity Act (ECOA) and the Fair Housing Act (FHA).
- Implemented a range of social science modelling techniques (e.g., ordinary least squares, logistic regression, and TOBIT functional forms) to predict consumer lending outcomes and evaluate ECOA / FHA violation risk.
- Organized large loan/application-level data files across Excel, SPSS, and Stata.
- Prepared materials for inclusion in professional reports and other work products delivered to clients.

2022 ***Intern****:* THG Geophysics, Corona, NM

- Acquired electrical resistivity profiles to assist with mapping karst hazards for potential wind farm locations.

RELEVANT SKILLS

Computer programming and data management: Python/JupyterLab, R/RStudio, ArcMap, ArcGIS Pro, MariaDB/MySQL, Grid Analysis and Display System (GrADS), SPSS, Stata, MatLab, Linux, UNIX, Adobe Illustrator, Microsoft Office Suite, Google Sheets.
Statistical and machine learning modeling: Model training, validation, and testing with XGBRegressor/RandomForestRegressor/PyTorch, linear regression, and logistic regression.
Lab instruments: Operation and calibration of Isoprime 100 isotope ratio mass spectrometer; in-class experience with petrographic microscope, laser diffraction particle size analyzer, X-ray diffractometer, scanning electron microscope.
Lab equipment and techniques: Experience with Micromill for speleothem drilling, intra-ring tree sampling, wet-sieving, loss-on-ignition analysis, and sample preparation for radiogenic isotope analysis.
Field Experience: Extraction of tree cores from deciduous and pine trees, collection of speleothems while spelunking, electrical resistivity testing.

TEACHING/MENTORSHIP EXPERIENCE

2022-2023 ***Undergraduate Research Opportunity Mentor**:* University of New Mexico

- Mentored an undergraduate student with a Python-based project to analyze drought quantification techniques.

2022-2023 ***Teaching Assistant**:* University of New Mexico

- ENVS 320L – Environmental Systems: Assisted students with ArcMap and Excel applications; taught lessons to reinforce understanding of key statistical and scientific concepts; aided in lab design; graded and provided feedback on assignments.
- GEOL 1110 – Introduction to Physical Geology: Grades and provides feedback on assignments; assists with assignment design.

LEADERSHIP EXPERIENCE

2024-Present ***United Graduate Workers, The Union for Everyone***, University of New Mexico

- Data Chair (2024-Present): Manages union membership salary databases, performs salary and membership data analysis tasks, leads UGW Data Team, advises executive council during weekly meetings, participates in collective bargaining negotiations.

2020–2021 ***Geology Club***, James Madison University

- Treasurer (2020–2021): Organized fundraisers; oversaw budgeting and banking; recorded apparel payments.
- Participated in community outreach at Massanutten Regional Library to educate young children about geology.

2019–2021 ***Sigma Gamma Epsilon Eta Nu Chapter***, James Madison University

- Treasurer (2020–2021): Oversaw budgeting, banking, and disbursement of funds.
- Secretary (2019–2020): Distributed emails, recorded meeting minutes, and reported club membership to Sigma Gamma Epsilon National Office.

2018–2021 ***Club Table Tennis***, James Madison University

- Table Tennis Chair (2018, 2020–2021): Led bi-weekly practices, directed in-house tournaments, and increased club's overall competitiveness.
- Vice President (2019): Led bi-weekly practices, supervised Table Tennis Chair and Secretary, and assisted the President with administrative tasks.

NOTABLE COURSEWORK

Graduate: Applied Machine Learning, Introduction to Machine Water Resource Modeling, Computational Methods for Geoscience Data with Python, Climate Dynamics, Spatial Statistics with R, Introduction/Advanced Programming for GIS with Python, Radiogenic Isotope Geochemistry, Geochemistry of Natural Waters

Undergraduate: Paleoclimatology and Paleoceanography**, Quantitative Methods in Geology**, Geochemistry of Natural Waters**, Hydrogeology, Advanced Stratigraphy, Structure, and Tectonics, Earth Surface Processes, Statistical Computing, Applied Linear Regression, Analysis of Variance and Experimental Design, Calculus I-III, Chemistry I-II w/ Labs, Physics I-II w/ Labs, Introduction to Programming with Python

***Additional work completed as part of the Honors Option curriculum, typically as independent research*